# Simulation Boundary Interaction Module (SBIM)

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## Governed Space

## Simulation Boundary Interaction

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## Canonical Definition

Simulation Boundary Interaction Module (SBIM) defines the governance framework for identifying, defining, controlling, documenting, and reconstructing the interaction boundaries through which a governed Simulation Model exchanges information, state, signals, actions, resources, events, or influence with the Simulation Environment, external systems, human participants, physical components, other models, or other simulation-relevant entities.

SBIM establishes the governed interaction layer between what belongs inside the Simulation Model, what belongs outside the Simulation Model, and what may legitimately cross the boundary between them.

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## Operational Role

SBIM is the third internal governance space of the Simulation Model & Environment Standard (SMES).

The preceding modules establish:

SMMM — What is the Simulation Model structurally composed of?

SECM — Within what configured Simulation Environment does that model operate?

SBIM asks:

Where do the model and its surrounding environment meet, what may cross that boundary, in which direction, under what interaction conditions, and how can those interactions be reconstructed?

The governed progression is:

Governed Model Structure → Governed Environment Configuration → Boundary Identification → Interaction Point Definition → Exchange Classification → Direction & Control Definition → Boundary Condition Establishment → Interaction Traceability → Governed Simulation Boundary Interaction

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## Module Operational Space

SBIM governs: Simulation Boundary identification, Model–Environment boundaries, model-to-model boundaries, simulation-to-external-system boundaries, virtual-to-physical boundaries, human-to-simulation boundaries, simulation-to-human boundaries, boundary interaction points, boundary interfaces, boundary channels, inbound interactions, outbound interactions, bidirectional interactions, exchanged information, exchanged state, exchanged signals, exchanged events, exchanged actions, exchanged resources, interaction direction, interaction timing, interaction sequencing, interaction frequency, interaction conditions, interaction permissions, interaction constraints, interaction transformations, interaction mediation, boundary synchronization, boundary failures, boundary interruption, boundary substitution, boundary change, boundary traceability, and boundary interaction reconstruction.

## Module Operational Scope

SBIM applies where Simulation Model behavior depends upon interaction across an identifiable simulation boundary.

This may include: fully virtual simulations, digital twins, software-in-the-loop simulations, hardware-in-the-loop simulations, human-in-the-loop simulations, autonomous-system simulations, robotics simulations, multi-agent simulations, distributed simulations, federated simulations, cyber-physical simulations, industrial simulations, infrastructure simulations, scientific simulations, financial simulations, transportation simulations, aerospace simulations, defense simulations, training environments, live-system-connected simulations, and future hybrid simulation environments.

Governance depth should remain proportional to the material influence of boundary interaction upon simulation behavior and intended simulation reliance.

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## Module Function

A Simulation Model and Simulation Environment may both be correctly defined while their interaction remains poorly governed.

A simulation can fail methodologically because: the wrong information enters the model, outputs cross the wrong interface, timing differs from intended conditions, a physical signal is transformed incorrectly, an external service behaves differently, a human intervention is not visible, two models exchange incompatible states, a boundary becomes unintentionally permeable, or an assumed interaction never actually occurs.

SBIM exists to make these interactions explicit.

Its function is to determine:

where interaction occurs,

what crosses the boundary,

in which direction,

through which mechanism,

under which conditions,

and:

what happens when the interaction changes or fails.

## Simulation Boundary

A Simulation Boundary is a governed separation between two simulation-relevant domains whose interaction may materially affect simulation behavior.

Examples include:

## Simulation Model ↔ Simulation Environment

## Model A ↔ Model B

## Simulation ↔ External Data Source

## Simulation ↔ Physical Hardware

## Simulation ↔ Human Operator

## Simulation ↔ Live Operational System

A boundary is not merely a technical interface.

It is the point at which one governed simulation domain may influence another.

## Boundary Interaction

A Boundary Interaction occurs when information, state, signals, events, actions, resources, control, or other material influence crosses a Simulation Boundary.

Conceptually:

## Domain A → Boundary → Domain B

or:

## Domain A ↔ Boundary ↔ Domain B Boundary Identification

Material Simulation Boundaries should be explicitly identifiable.

Where relevant, boundary identification should establish: boundary identity, connected domains, boundary type, interaction purpose, direction, applicable interface, exchanged object or signal, interaction conditions, and applicable configuration.

## Boundary Type

A Simulation Boundary may be classified according to the domains it separates.

Possible types include:

## Model–Environment Boundary

## Model–Model Boundary

## Simulation–External System Boundary

## Simulation–Physical System Boundary

## Simulation–Human Boundary

## Simulation–Operational Reality Boundary

## Simulation–Data Source Boundary

## Simulation–Service Boundary

or another domain-specific boundary.

## Interaction Point

An Interaction Point is the identifiable location, mechanism, interface, event, or condition through which a boundary exchange occurs.

Examples include: API endpoints, message queues, sensor interfaces, actuator interfaces, network ports, shared state objects, data feeds, event buses, human controls, hardware interfaces, model coupling interfaces, or physical measurement channels.

## Boundary Interface

A Boundary Interface should remain distinguishable from the boundary itself.

The boundary establishes separation.

The interface provides a mechanism through which governed interaction may occur.

Therefore:

## Boundary ≠ Interface

and:

Interface ≠ Permission for unrestricted interaction.

## Inbound Interaction

An Inbound Interaction introduces information, state, events, signals, resources, commands, or influence into the governed Simulation Model or Simulation Environment.

Conceptually:

External Domain → Simulation Boundary → Governed Simulation Domain Outbound Interaction

An Outbound Interaction transfers simulation-generated information, state, events, signals, actions, or influence beyond the governed domain.

Conceptually:

Governed Simulation Domain → Simulation Boundary → External Domain Bidirectional Interaction

Some boundaries permit continuous or event-driven exchange in both directions.

Conceptually:

## Domain A ↔ Domain B

Bidirectionality should remain explicit because feedback can materially alter simulation behavior.

## Interaction Direction

Interaction direction should be defined where material.

Possible states include:

Inbound

Outbound

## Bidirectional

Conditional

or:

Dynamic

where direction changes according to governed simulation conditions.

## Interaction Object

An Interaction Object is whatever materially crosses the boundary.

It may include: data, state, command, event, signal, measurement, resource, control instruction, model output, environmental response, human input, or physical response.

## Interaction Classification

Material interactions may be classified according to their role.

For example:

## Informational

## State-Changing

Control

Resource

## Observational

Triggering

Feedback

## Synchronization

or:

## Safety-Critical

Classification should support the level of governance required by the simulation inquiry.

## Interaction Conditions

A boundary interaction may occur only under defined conditions.

These may include: model state, environment state, scenario state, timing condition, event trigger, threshold, permission state, resource availability, synchronization condition, or external-system availability.

## Boundary Interaction Rule

A Boundary Interaction Rule establishes the conditions under which an interaction may occur and the expected behavior of the interaction mechanism.

Conceptually:

Trigger + Boundary Condition + Interaction Object + Direction + Transformation + Destination Interaction Timing

Timing may materially affect simulation behavior.

SBIM therefore governs relevant properties such as: interaction time, delay, latency, timeout, frequency, sequence, synchronization, asynchronous behavior, buffering, and timing uncertainty.

## Interaction Sequence

Some boundary interactions must occur in a defined sequence.

For example:

## Request → Authorization → Exchange → Response → State Update

Changing the sequence may materially alter simulation behavior even when the same information is exchanged.

## Interaction Frequency

Interaction frequency may be:

Continuous

Periodic

## Event-Driven

On-Demand

## Single-Occurrence

or:

Adaptive

where frequency changes according to simulation state.

## Boundary Synchronization

Where multiple simulation domains operate under different clocks, execution rates, or event systems, boundary synchronization should be governed.

This may include: time synchronization, state synchronization, event synchronization, step synchronization, clock conversion, or synchronization tolerance.

## Synchronization Failure

A technically functioning interface may still produce invalid simulation behavior if the participating domains are not sufficiently synchronized.

SBIM therefore treats synchronization as a boundary-governance question where it materially affects interaction.

## Interaction Transformation

Information crossing a boundary may be transformed.

Examples include:

## Unit Conversion

## Coordinate Transformation

## Protocol Conversion

## Data Encoding

## Signal Scaling

## State Mapping

Aggregation

Filtering

## Normalization

or:

Abstraction

Material transformations should remain identifiable.

## Transformation Traceability

Where a transformation materially affects meaning or behavior, SBIM should preserve:

Source Interaction Object → Transformation → Resulting Interaction Object No Invisible Transformation Principle

A material transformation must not alter what crosses a Simulation Boundary without remaining visible to simulation governance.

## Interaction Mediation

Some interactions are mediated by another component.

Conceptually:

## Model → Middleware → Environment

or:

## Physical Sensor → Interface Adapter → Simulation

The mediator should remain identifiable where its behavior may materially affect the interaction.

## External Service Interaction

A Simulation Model may rely upon external services.

Examples include: external APIs, mapping services, weather services, AI models, databases, cloud services, communication services, or live operational feeds.

SBIM governs the interaction boundary.

SECM governs the environmental dependency where the service forms part of the Simulation Environment configuration.

## Model-to-Model Interaction

Where multiple models interact, SBIM governs the exchange relationship between them.

For example:

## Vehicle Dynamics Model ↔ Control Model

or:

## Economic Model ↔ Population Model

SMMM governs their structural composition.

SBIM governs what crosses the boundary separating them.

## Federated Simulation Interaction

In distributed or federated simulations, independent simulation components may operate under separate execution contexts.

SBIM should preserve sufficient governance over: participant identity, interaction boundaries, exchanged state, timing, synchronization, transformation, and interaction dependencies.

## Human-to-Simulation Interaction

Human participants may materially influence simulation behavior through: commands, decisions, manual overrides, classifications, scenario interventions, control inputs, or observational feedback.

Such interaction should remain visible where material.

## Simulation-to-Human Interaction

The simulation may also influence human behavior through: displays, alerts, recommendations, control interfaces, training feedback, or decision-support outputs.

SBIM governs the simulation interaction boundary, not the broader governance of human cognition or decision accountability.

## Hardware-in-the-Loop Boundary

Hardware-in-the-loop simulations create a boundary between:

## Simulated Domain ↔ Physical Hardware

The boundary may involve: physical signals, sensor emulation, actuator commands, timing, conversion, communication, and hardware response.

These interactions should remain sufficiently reconstructable to determine what the simulation and physical component actually exchanged.

## Software-in-the-Loop Boundary

Software-in-the-loop environments may connect:

## Simulation Environment ↔ Operational Software Component

The operational software may behave differently from a simplified simulated component.

The boundary should therefore preserve the identity and configuration of the interacting software where material.

## Live-System Interaction

Some simulations interact with live operational systems.

SBIM requires clear separation between:

## Simulated State

and:

## Operational State

where the distinction matters.

Simulation-generated actions should not silently enter Operational Reality merely because a technical connection exists.

## Simulation-to-Reality Boundary

The boundary between simulation and actual Operational Reality is especially significant.

SIMULOS® governs how the simulation interacts with that boundary.

ORA™ governs the actual Operational Reality beyond it.

Conceptually:

## Simulation Domain → Governed Simulation Boundary → Operational Reality

The architectures remain adjacent rather than duplicative.

## Boundary Permission

Technical connectivity does not itself establish legitimate interaction.

Where relevant, the simulation should identify whether an interaction is:

Permitted

## Conditionally Permitted

Restricted

Prohibited

or:

## Undetermined

within the governed simulation configuration.

## Boundary Constraint

A Boundary Constraint limits how interaction may occur.

Examples include: permitted data types, maximum signal range, frequency limits, latency limits, directional restrictions, state restrictions, rate limits, prohibited commands, or interaction-duration limits.

## Boundary Condition

A Boundary Condition establishes the governed conditions under which the behavior of the model at a simulation boundary is defined.

Boundary conditions may be: fixed, dynamic, stochastic, scenario-dependent, state-dependent, time-dependent, or externally supplied.

## Boundary Condition ≠ Simulation Parameterization

SBIM governs the existence and interaction role of boundary conditions.

SPHM governs the broader controlled parameterization of the Simulation Model and Environment.

This separation prevents duplication between the two modules.

## Boundary Failure

A Boundary Failure occurs when an intended interaction cannot occur or occurs outside its governed conditions.

Examples include: lost communication, malformed exchange, interface failure, timing failure, synchronization failure, unavailable external service, missing response, incorrect transformation, or prohibited interaction.

## Boundary Failure Behavior

Where material, the simulation should define what happens when boundary interaction fails.

Possible responses include:

Retry

Fallback

Hold State

## Substitute Value

## Terminate Interaction

## Terminate Simulation

or another governed response.

## Boundary Substitution

A failed or unavailable boundary interaction may be replaced by another mechanism.

For example:

## Live Weather Feed → Synthetic Weather Generator

Such substitution may materially change the simulation context.

It should therefore remain visible.

## Interaction Interruption

Temporary interruption should be distinguishable from permanent boundary failure where the difference affects simulation behavior.

## Boundary Change

A boundary may change through: interface replacement, interaction-object change, protocol change, timing change, transformation change, permission change, connected-domain change, synchronization change, or interaction-rule change.

Material boundary changes should remain identifiable.

## Boundary Drift

Boundary Drift occurs where actual interaction behavior progressively differs from the governed boundary definition without an equivalent governed change.

Examples include: undocumented API changes, increasing latency, new message fields, changed transformation logic, changed external service behavior, altered synchronization, or unintended new interaction channels.

## No Hidden Boundary Principle

A material interaction boundary must not remain outside simulation governance merely because the exchange is technically automatic, externally managed, embedded, inherited, or assumed to be stable.

## No Hidden Interaction Principle

A material influence crossing a Simulation Boundary must not affect simulation behavior without remaining identifiable as part of the governed interaction structure.

## No Boundary Bypass Principle

A governed Simulation Boundary must not be bypassed through an alternative interaction path that escapes the controls, conditions, or traceability established for that boundary.

## No Simulation-Reality Leakage Principle

Simulation-generated state, commands, or actions must not silently become operational actions merely because the simulation is technically connected to a live environment.

## Boundary Equivalence

Two boundary configurations should not be treated as equivalent merely because they connect the same domains.

Material differences may include: protocol, latency, transformation, direction, frequency, synchronization, permission, or failure behavior.

## Boundary Interaction Completeness

Boundary Interaction Completeness asks:

Have all material interaction pathways capable of influencing the governed simulation inquiry been identified?

A technically undocumented pathway may still be governance-relevant.

## Boundary Interaction Consistency

Boundary Interaction Consistency asks whether the defined interactions remain compatible with: Model Structure, Environment Configuration, Simulation Purpose & Scope, Reality Representation, and Simulation Assumptions.

## Boundary Interaction Coherence

An interaction may be technically valid while still being incoherent with the simulation design.

For example, correct data delivered at the wrong simulation time may create behavior inconsistent with the intended environment.

SBIM therefore distinguishes:

## Technical Connectivity

from:

Governed Interaction Coherence.

## Boundary Interaction Change Propagation

A material boundary change may propagate into later SIMULOS® governance spaces.

Conceptually:

## Boundary Change

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## Parameter / Input Effect

↓

## Scenario Effect

↓

## Execution Effect

↓

## Output Effect

↓

## Reality-Correlation Effect

SBIM identifies and preserves the boundary change.

Affected downstream spaces govern their respective consequences.

## Interaction Binding

Material boundary interactions should be bindable to the applicable: Simulation Model version, Simulation Environment configuration, boundary configuration, and simulation execution.

Conceptually:

Simulation Execution → Model Version + Environment Configuration + Boundary Interaction Configuration Boundary Interaction Reconstruction

SBIM should enable reconstruction sufficient to answer:

Which boundaries existed?

Which domains did they connect?

What crossed each boundary?

In which direction?

Through which interface?

Under which conditions?

Which transformations occurred?

Which timing and synchronization rules applied?

Did any boundary fail, change, or become substituted?

## Boundary Interaction Traceability

A complete traceability path may be represented as:

## Simulation Model

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## Simulation Environment

↓

## Simulation Boundary

↓

## Interaction Point

↓

## Interaction Object

↓

Direction

↓

## Transformation / Mediation

↓

## Boundary Conditions

↓

## Interaction Event

↓

## Simulation Execution

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## Minimum Implementation Framework

### 1. Identify Material Simulation Boundaries

Determine where the Simulation Model interacts with the environment, other models, humans, hardware, services, or external systems. 2. Define Boundary Interaction Points

Identify the mechanisms through which material interaction occurs. 3. Classify Boundary Exchanges

Determine what crosses each boundary and its simulation role. 4. Establish Direction and Conditions

Define interaction direction, timing, triggers, constraints, and applicable conditions. 5. Govern Transformation and Mediation

Preserve material transformations and intermediary mechanisms. 6. Establish Synchronization

Define timing and state synchronization where interaction depends upon coordinated execution. 7. Establish Boundary Failure Behavior

Determine how interruption, failure, substitution, and unavailable interactions are handled. 8. Bind Boundaries to Model and Environment Configuration

Ensure the interaction configuration corresponds to the governed model and environment. 9. Control Material Boundary Change

Preserve changes capable of affecting simulation behavior. 10. Preserve Interaction Reconstruction

Maintain sufficient traceability to reconstruct material boundary interactions associated with simulation execution.

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## Governance Outputs

SBIM may produce: Simulation Boundary Register, Simulation Boundary Identifier, Boundary Type Record, Boundary Interaction Map, Interaction Point Register, Boundary Interface Record, Interaction Object Record, Interaction Classification Record, Interaction Direction Record, Interaction Condition Record, Boundary Interaction Rule, Interaction Timing Record, Interaction Sequence Record, Interaction Frequency Record, Boundary Synchronization Record, Interaction Transformation Record, Transformation Traceability Record, Interaction Mediation Record, External Service Interaction Record, Model-to-Model Interaction Record, Human-Simulation Interaction Record, Hardware-in-the-Loop Boundary Record, Software-in-the-Loop Boundary Record, Live-System Interaction Record, Boundary Permission Record, Boundary Constraint Record, Boundary Condition Record, Boundary Failure Record, Boundary Substitution Record, Boundary Change Record, Boundary Drift Record, Boundary Interaction Completeness Assessment, Boundary Interaction Consistency Assessment, Boundary Interaction Coherence Assessment, Boundary Interaction Configuration, Boundary Interaction Reconstruction Record, and Simulation Boundary Interaction Traceability Map.

## Use Case 1 — Autonomous Vehicle Hardware-in-the-Loop Simulation

### Scenario

An autonomous driving simulation connects a virtual road environment to a physical vehicle controller.

The boundary carries:

## Virtual Sensor Signals → Physical Controller

and:

# Controller Commands → Virtual Vehicle Model Application

SBIM governs: signal direction, timing, conversion, synchronization, interface conditions, and failure behavior.

A 100 ms additional interface delay cannot remain invisible if it materially changes controller behavior. Result

The simulation execution can later demonstrate exactly how the virtual environment and physical controller interacted. Use Case 2 — Multi-Model Space Mission Simulation

### Scenario

A mission simulation couples:

Orbital Model ↔ Guidance Model ↔ Communication Model ↔ Ground-System Model Application

SBIM establishes the boundaries between the models and governs: exchanged states, message direction, timing, synchronization, transformations, and failure behavior.

### Result

If the communication model becomes unavailable during a simulation run, the resulting substitution or interruption remains visible rather than becoming an unexplained change in mission behavior. Use Case 3 — Digital Twin Connected to Operational Reality

### Scenario

A digital twin receives live sensor data from an industrial facility and may generate recommendations for operators. Application

SBIM separates:

# Operational Sensor Data → Simulation

from:

# Simulation Recommendation → Human Operator

and prevents the latter from being treated as direct operational control unless such interaction is explicitly governed. Result

The simulation-to-reality boundary remains explicit, preserving the distinction between simulated state, operational state, and any permitted interaction between them.

# Architectural Position

Simulation Boundary Interaction is the third internal governance space of the Simulation Model & Environment Standard (SMES).

The complete SMES progression is:

Simulation Model Structure → Simulation Environment Configuration → Simulation Boundary Interaction → Simulation Parameterization → Simulation Environment Versioning

SBIM receives the governed model structure from SMMM and the governed environment configuration from SECM.

It establishes the interaction layer required before the model and environment can be parameterized as a controlled simulation configuration.

Within the broader SIMULOS® lifecycle:

Simulation Object → Simulation Purpose & Scope → Reality Representation → Simulation Assumptions → Simulation Model & Environment → downstream simulation governance

SBIM therefore governs the interfaces through which otherwise separately governed simulation domains become behaviorally connected.

## Foundational Principle

A simulation boundary is governable only when the material interactions crossing it can be identified, bounded, controlled, and reconstructed.

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## Canonical Closing Statement

Simulation Boundary Interaction Module establishes the interaction-boundary governance layer of the Simulation Model & Environment Standard by defining how Simulation Models, Simulation Environments, external systems, physical components, humans, services, and other models may exchange material information, state, signals, events, actions, resources, and influence.
Through boundary identification, interaction classification, direction, timing, synchronization, transformation, mediation, permission, constraints, failure behavior, substitution, change control, execution binding, and reconstruction, SBIM ensures that simulation behavior cannot be materially shaped by invisible or uncontrolled exchanges across the boundaries separating the governed components of the simulation.

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